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Hybrid data decomposition-based deep learning for Bitcoin prediction and algorithm trading

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In recent years, Bitcoin has received substantial attention as potentially high-earning investment. However, its volatile price movement exhibits great financial risks. Therefore, how to accurately predict and capture changing trends in the Bitcoin market is of substantial importance to investors and policy makers. However, empirical works in the Bitcoin forecasting and trading support systems are at an early stage. To fill this void, this study proposes a novel data decomposition-based hybrid bidirectional deep-learning model in forecasting the daily price change in the Bitcoin market and conducting algorithmic trading on the market. Two primary steps are involved in our methodology framework, namely, data decomposition for inner factors extraction and bidirectional deep learning for forecasting the Bitcoin price. Results demonstrate that the proposed model outperforms other benchmark models, including econometric models, machine-learning models, and deep-learning models. Furthermore, the proposed model achieved higher investment returns than all benchmark models and the buy-and-hold strategy in a trading simulation. The robustness of the model is verified through multiple forecasting periods and testing

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Introduction

As the price of Bitcoin increased from almost zero in 2009 to nearly \$20,000 in 2017, it attracted considerable attention from investors and policy makers. This significant increase in its price was accompanied by a steady growth in the Bitcoin market; as of December 2019, the average market volume is approximately \$19.45 billion per day. Acting as a currency, Bitcoin has several characteristics: decentralized transaction, auditability and anonymity.

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emissions created from the mining of Bitcoin could increase the global temperature by 2 °C. However, despite the scholarly attention, empirical works on Bitcoin pricing forecasting models remain at a relatively early stage.

Some natural questions arise regarding Bitcoin price prediction: Is Bitcoin price predictable? What factors affect its market price? Potential factors affecting Bitcoin price movements discussed by previous studies can be classified into two groups: The first group comprises internal factors, which includes features, such as Bitcoin market volume, Bitcoin transactions patterns, and hash rate. For example, Chen et al. ([2019](#)) reported that Bitcoin network, Bitcoin trading, investor attention, and gold price can effectively influence Bitcoin price. Jang and Lee ([2018](#)) indicate that factors, such as the hash rate, difficulty, and block size, can cause Bitcoin price movement. Catania et al. ([2019](#)) applied a set of cryptopredictors to study the predictability of a cryptocurrency time series. The second group involves external factors, which include features, such as external commodities, investor sentiment,

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models also have a major drawback in that the models all assume the time series are linear and stationary, which is hardly satisfied by the volatile and non-stationary nature of the Bitcoin market (Yu et al. [2008](#)). As a result, they are less effective in predicting the Bitcoin market price.

Some studies have adopted machine-learning methods to develop prediction models for the financial market, which attempt to capture the nonlinear characteristics of financial time series (Thakkar and Chaudhari [2020b](#), [2021a](#); Thakkar et al. [2021](#); Thakkar and Lohiya [2021](#)). Kristjanpoller and Minutolo ([2018](#)) propose a framework integrating GARCH and ANN to forecast the price volatility of Bitcoin. Peng et al. ([2018](#)) use support vector regression (SVR) to predict the volatility of cryptocurrencies. More recently, a few studies have adopted deep-learning models to forecast the financial market price as they have shown superior performance over their shallow counterparts (Ramadhani et al. 2018; Thakkar and Chaudhari [2020c](#); Thakkar and Chaudhari [2021b](#); Chaudhari and Thakkar [2021](#)). For

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mode decomposition (CEEMD) and extended extreme learning machine to forecast crude oil prices. Wen et al. ([2017](#)) use CEEMD and combined SVM and ANN to forecast gold prices. Santhosh et al. ([2019](#)) combine EEMD and Deep Boltzman Machines to forecast wind energy. However, the prediction error from the individual decomposed modes tend to accumulate, which could negatively affect the forecasting results of the prediction model (Tang et al. [2015](#)). Moreover, a mode-mixing problem may occur in the process of EMD, which can produce oscillations with similar scales in IMF (Colominas et al. [2014](#)).

Based on previous studies discussed above, this paper proposes a novel data decomposition-based hybrid bidirectional deep-learning model to forecast the Bitcoin market price. First, a non-recursive signal decomposition method, variational mode decomposition (VMD), is introduced to decompose historical Bitcoin price data into various intrinsic modes. In comparison to the widely adopted EMD method, VMD can effectively avoid the mode-mixing problem (Dragomiretskiy and Zosso [2014](#)). Second, a bidirectional long short-term memory

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approach, decomposes the original time series into different cycle factors and then integrates them to generate the final predictive output, which can effectively improve the prediction accuracies on non-stationary time series.

Our intended contributions of this paper may lie with our proposal of a synthetic framework of Bitcoin price prediction incorporating influential factors of macroeconomics and investor behaviors based on data decomposition and deep-learning approaches. In particular, we incorporate Google trends in order to utilize the hidden and effective information of irrational behaviors in the Bitcoin market to improve the forecasting results. We empirically confirm the effectiveness of our proposed hybrid deep-learning models for Bitcoin price forecasting. Our proposed model outperforms several benchmark econometrics, machine-learning models, deep-learning models, and hybrid learning models. In addition, we extend the practical implications of our paper to conduct algorithmic trading based on the forecasting model. The methodology and empirical results of our study shed new light on

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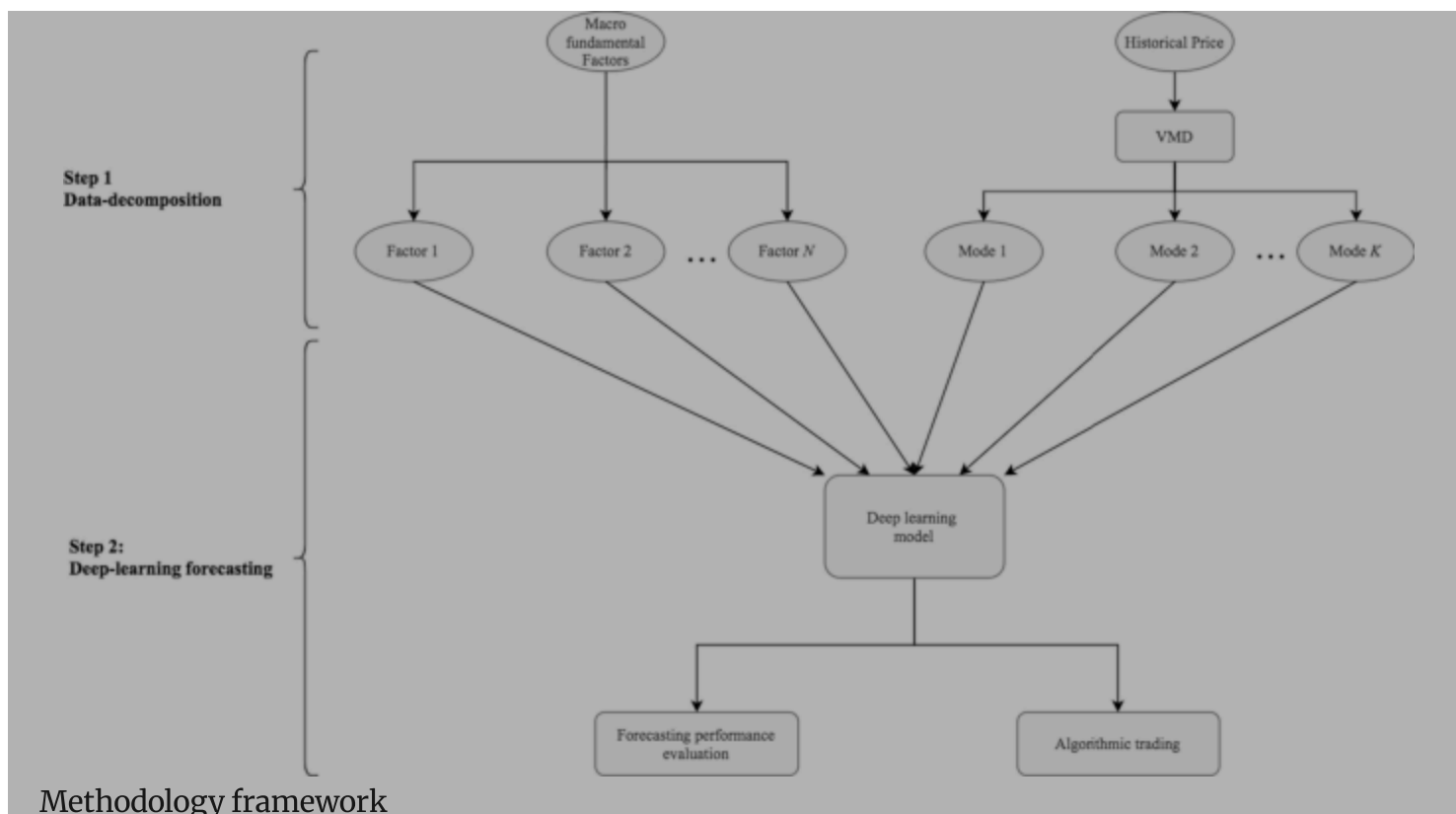
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Methodology framework

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“[Variational mode decomposition](#)” and “[Bidirectional LSTM](#)” sections provide a detailed description into the corresponding techniques of VMD and bidirectional LSTM, respectively.

Variational mode decomposition

VMD is an entirely non-recursive signal decomposition technique proposed by Dragomiretskiy and Zosso ([2014](#)). Based on Wiener filtering and Hilbert transform (Wang and Markert [2015](#)), it decomposes the original input signal $(f(t))$ into a series of quasi-orthogonal band-limited discrete sub-signals $(\{u_k\})$ that are mostly centered tightly around their respective center frequency $(\{\omega_k\})$ (Li et al. [2021](#)). In essence, VMD is a variational optimization problem that seeks to minimize the total bandwidth of each mode. The optimization procedure is as follows (Zhang et al. [2017](#)):

Step 1: Calculate the Hilbert transform of each mode $(\{u_k\})$ and transform into respective uni-sided frequency spectrum:

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Dirac delta function, $(\otimes)$ denotes the convolution operator, and $(f(t))$ represents the original input signal.

To obtain the optimal solution of the constrained optimization problem in Eq. (1), a quadratic penalty function (α) and a Lagrangian multiplier (λ) are introduced for finite convergence and constraint enforcement purposes. Thus, the augmented Lagrangian multiplier function (L) can be obtained as follows:

$$\begin{aligned} L & \left(\left\{ u_k \right\}, \left\{ \omega_k \right\}, \right. \\ & \left. \lambda \right) = \alpha \left\{ \sum_{k=1}^K \left| \left(\left(\delta(t) + \frac{j}{\pi t} \right) \otimes u_k \right) \right| e_k^2 - j \omega_k t \right\} \\ & + f(t) - \sum_{k=1}^K u_k \left(t \right) \\ & + \lambda \left(t \right), f(t) - \sum_{k=1}^K u_k \left(t \right) \end{aligned}$$

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The optimal solution is then obtained using the alternative direction method of multipliers, and the original input signal $f(t)$ is decomposed into K sub-signal modes.

Bidirectional LSTM

Proposed by Schuster and Paliwal (1997), the bidirectional recurrent neural network is a recurrent neural network (RNN) that utilizes both forward and backward information in the data. In this paper, the traditional RNN cells are replaced by LSTM cells. A LSTM cell consists of an input gate i_t , a forget gate f_t , an output gate o_t , as well as a memory cell block C_t . The forget gate f_t and input gate i_t are defined as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

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$$h_t = o_t * \tanh(\{C_t\})$$

(10)

where $\{x_t\}$ denotes the input at time t . σ represents the sigmoid function and $*$ is the element-wise multiplication. W and b are the respective weight matrices and bias vectors.

As illustrated in Fig. 2, the BiLSTM contains two hidden layers, where one of the layers processes information in the forward direction and the other layer processes information in the reverse direction. The two hidden layers are connected to one output layer so that the BiLSTM neural network can learn the information from two different data directions. Since time-series data contains a two-way sequential relationship as the current state is not only the reflection of historical information but also the basis of the future state. Therefore,

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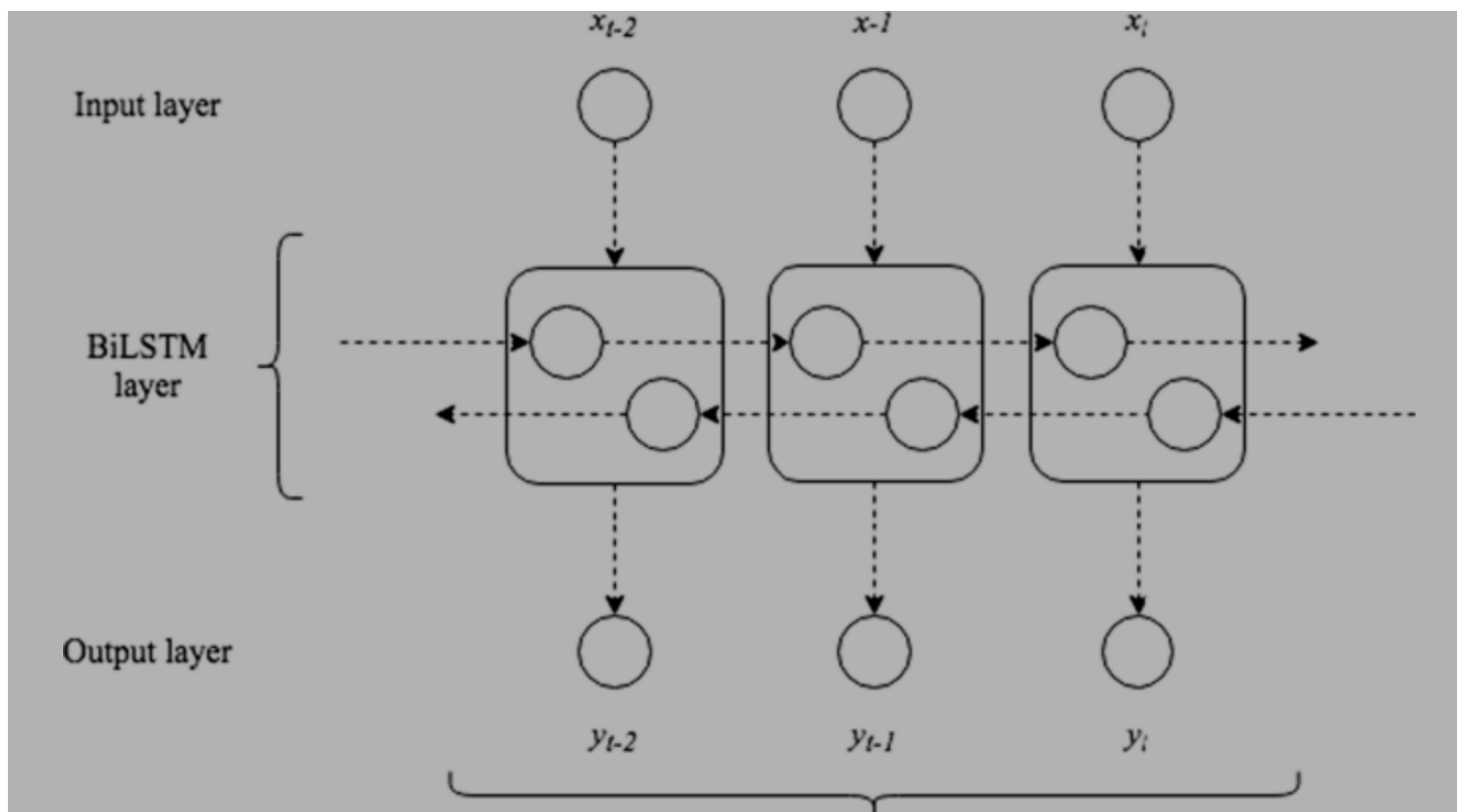
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“[Experiment design](#)” section provides a detailed description about the experimental design. “[Empirical results](#)” section presents the results and verifies that the proposed model is robust across different market conditions and forecasting horizons.

Experiment design

Data description

This study aims to forecast the price fluctuations in the Bitcoin market and conduct algorithmic trading based on the prediction results. The data used in this study consist of the historical daily Bitcoin market closing price time series obtained from Quandl (www.quandl.com). The raw data is from the period of April 29, 2013 to January 1, 2021, with a total of 2805 observations. A graphical representation of the data is illustrated in Fig. [3](#). Table [1](#) presents the common descriptive statistics for the daily Bitcoin market price, while also introducing the augmented Dickey–Fuller test (ADF). The null hypothesis is rejected in

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Table 1 Common descriptive statistics of Bitcoin price

Table 2 Macroeconomical factors

The data are divided into two sets: a training set and a testing set. The preceding 90% of the data are used to train the prediction model and the remaining 10% are to evaluate the model performance. Overall, the training set consists of 2525 observations from April 29, 2013 to March 27, 2020. The testing set contains 280 observations from March 28, 2020 to January 1, 2021.

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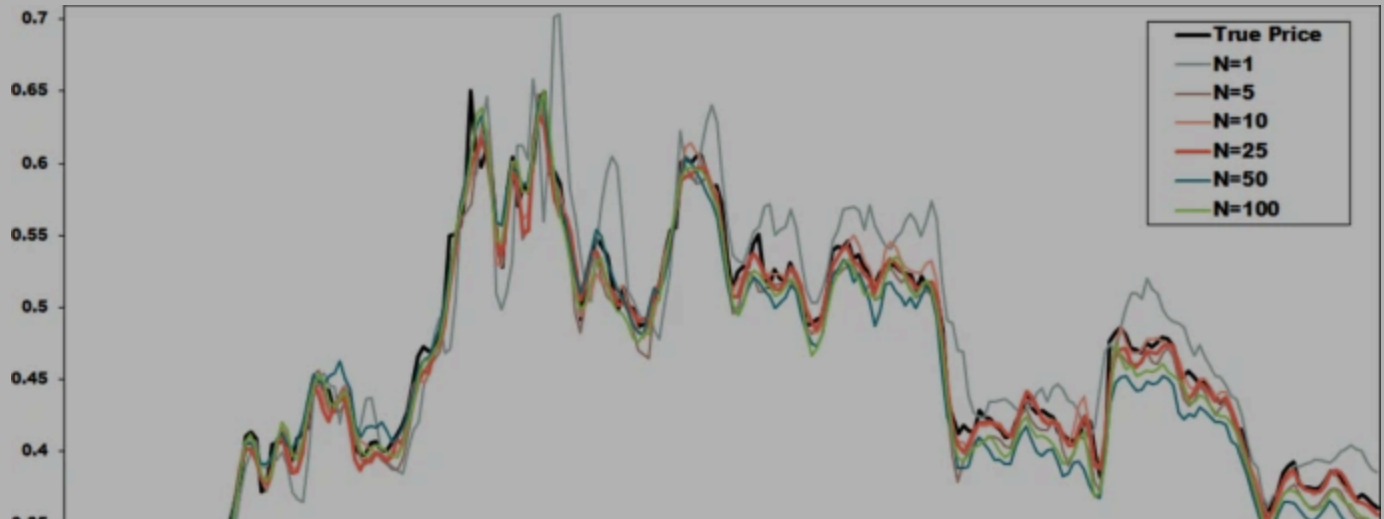
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when $(N=25)$, the model yielded the best prediction performance. As a result, the sliding-window length for the model is set to $(N=25)$.

Fig. 4



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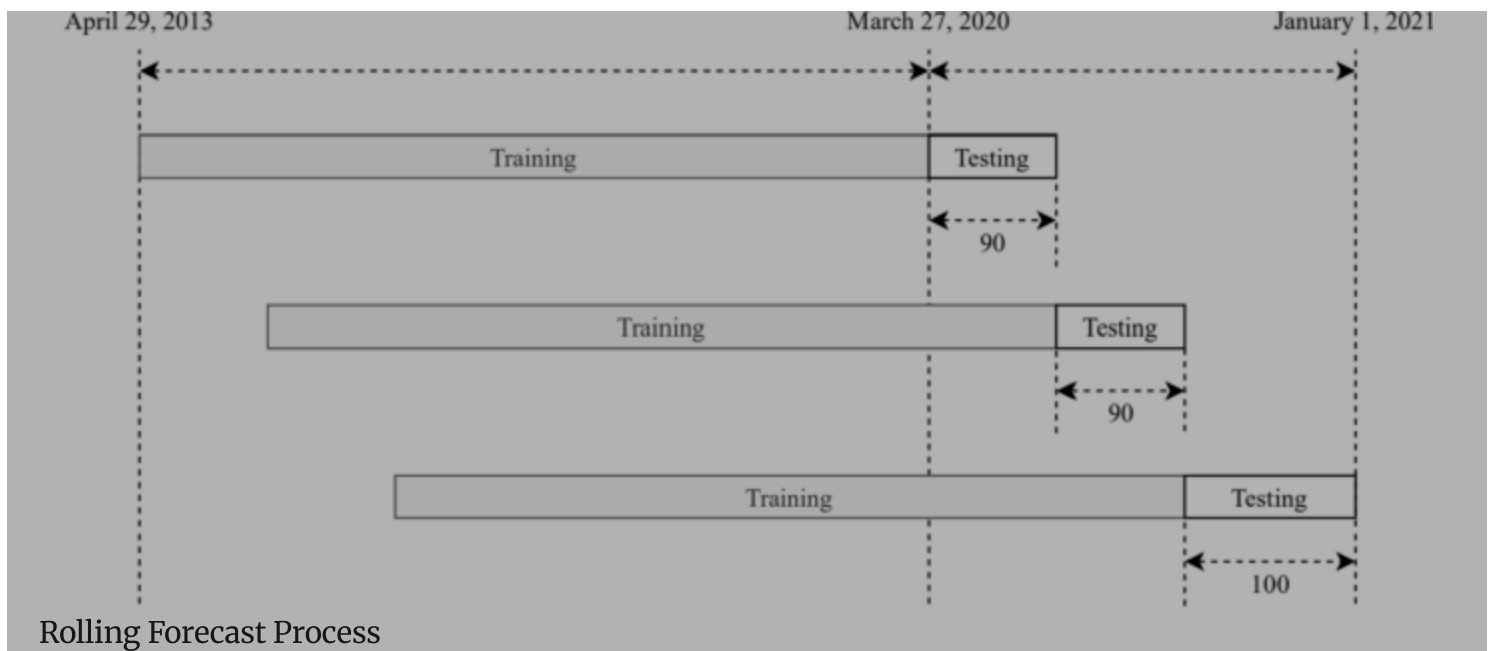
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$$MAE = \frac{1}{N} \sum_{t=1}^N |\hat{x}_t - x_t|$$

(15)

where $\hat{x}_t$ and x_t , $t=1, 2, \dots, N$ are the predicted value and the actual value at time t , and N represents the total number of data points in the testing set.

Moreover, directional accuracy (DA) is introduced to assess the market trend predictive ability of the model (Yu et al. [2008](#)). The larger the DA, the better the model market trend predictive ability:

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can comprehensively assess the effectiveness of signal decomposition technique in improving the Bitcoin market forecasting performance.

Second, in order to assess the effectiveness of different decomposed inner factors in improving the forecasting performance of the model, hybrid one- and two-characteristic models are formulated by importing factors of different frequencies into the proposed model.

The different inner factors extracted through data decomposition are classified as low-frequency, medium-frequency, and high-frequency modes based on their respective periodicity. They are selected respectively as input features to construct the corresponding hybrid one-characteristic models: VMD-L-BiLSTM, VMD-M-BiLSTM, and VMD-H-BiLSTM. For hybrid two-characteristic models, modes from two different frequencies are imported as input features, which results in three models: VMD-LM-BiLSTM, VMD-LH-BiLSTM, and VMD-MH-BiLSTM. For each benchmark model, the parameters and lag-order

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having the highest. The decomposed modes contain different inner factors hidden in the original signal that have various effects on the price movement in the Bitcoin market.

Based on the detected cyclicity as shown in Fig. 6, each mode is classified into one of three groups: low frequency, medium frequency and high frequency (Zhu et al. 2019; Li et al. 2021). In terms of Bitcoin market price, M1 has a cycle of approximately three years, which is significantly longer than other decomposed modes. Thus, it is classified as the low-frequency mode, which captures the long-term trend in the Bitcoin market prices from 2013 to 2019. Modes M2–M4 have cycles between three months and eight months, which are considered as the medium modes. The medium-frequency modes may represent price shocks brought on by economic and political events related to Bitcoin and other cryptocurrencies. Modes M5–M11 are regarded as high-frequency modes with relatively short cycles (4–21 days), which may possibly reflect the short-term fluctuations, such as investor speculations, that exist on the market.

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As can be seen from Table 3, the five single benchmark models and the proposed VMD-LMH-BiLSTM model displayed significantly different model performances. Looking at the levels of prediction accuracy between the single models, the BiLSTM model performed better than the traditional LSTM model. Taking a one-step-ahead prediction as an example, the $(MSE, RMSE, MAPE)$ and (MAE) criteria are decreased by 18.18%, 10.21%, 9.45% and 8.94%, respectively. This indicates that the bidirectional structure is superior to the traditional monodirectional neural networks structure. By adopting a bidirectional structure, the model is able to extract more information within the time series, thus yielding better performance results. When comparing the levels of prediction accuracy between the proposed model and the single models, the proposed VMD-LMH-BiLSTM model displayed far superior fitting performance by operating better across all criteria $(MSE, RMSE, MAPE)$ and (MAE) .

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Forecasting performance evaluation with hybrid benchmark models

Although the decomposed inner factors can significantly enhance the market prediction accuracy of the model, they have different cycles that range from several days to several years. The differences in period lengths may indicate that for short-term price predictions in the Bitcoin market, not all decomposed factors contribute equally to the improved predictive ability of the model. Therefore, we further assess the effectiveness of different decomposed inner factors in improving the forecasting performance of the model by comparing the performances of the proposed VMD-LMH-BiLSTM model with hybrid one- and two-characteristic models.

The proposed VMD-LMH-BiLSTM model and the six hybrid benchmark models are formulated and utilized to forecast the daily price of the Bitcoin market. The one-step prediction result comparisons and prediction accuracy results are illustrated in Table 4. According to the comparison results, the proposed VMD-LMH-BiLSTM model not only

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VMD-L-BiLSTM achieved the worst performance, resulting in a significant increase across all criteria ($(MSE, RMSE, MAPE)$ and (MAE)). These results indicate that the inner factors of different frequencies decomposed from the original Bitcoin market time series have various effects on the prediction model.

A similar pattern can be observed in the hybrid two-characteristic models. When the high-frequency modes are removed from the proposed VMD-LMH-BiLSTM model, the prediction performance of the resulting VMD-LM-BiLSTM model suffers a significant decrease, with the $(RMSE, MAPE)$ and (MAE) increasing by 4.93, 5.08, and 5.19 times, respectively. When the medium-frequency inner factors are removed, the $(RMSE)$ of the VMD-LH-BiLSTM model doubled from 0.007 to 0.014. By comparison, with the low-frequency modes removed, the VMD-MH-BiLSTM model achieved better prediction performance than the other two hybrid two-characteristic-models and obtained the smallest $(MSE, RMSE, MAPE)$ and (MAE) values. These further indicate that although the short-term Bitcoin

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The medium-frequency modes, which have periods ranging from one to eight months, represent economic and political events related to Bitcoin and other cryptocurrencies. These events, such as international regulatory policies, are important and influential factors on Bitcoin market price volatility over the medium-term. For example, the Bitcoin market prices crashed in early January, which corresponds to the time when multiple governments such as China, South Korea, and the United States announced tightened regulations on Bitcoin trading (Cumming et al. [2019](#)). Over time, the medium-frequency component reverts to the mean, which indicates that the Bitcoin market has absorbed the influences of these events. As a result, the Bitcoin market price eventually returned to its long-term trend.

The high-frequency inner factors have the shortest periods of all three components, which range from 4 to 21 days. These factors may possibly reflect short-term fluctuations such as investor speculations that exist on the market. These random disturbances exhibit mean-reversion characteristics with very short cycles, which indicate that their influences are

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Robustness check

In order to further evaluate the prediction performance and verify the robustness of the proposed approach, we employ the following two-step approach: First, we construct benchmark models using various state-of-the-art forecasting techniques to conduct horizontal performance comparison. For horizontal performance comparison, we construct four benchmark models, which utilize the same decomposition and sentiment input features as the proposed approach: econometric-based, traditional machine-learning, deep-learning, and hybrid approaches. The four benchmark models, in turn, include four models: ARIMA, LR, unidirectional LSTM, and EMD LSTM, which are utilized to predict Bitcoin price by previous works (Altan et al. [2019](#); Wirawan et al. [2019](#); Cohen [2020](#)). Second, since the trading rule of Bitcoin is $T + 0$ plus 24 h, we also apply our proposed approach to higher frequency price data and compare its forecasting performance against the four benchmark models horizontally.

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ARIMA model, LR model, unidirectional LSTM model, and EMD LSTM model. To ensure consistency, we utilize the same decomposition and sentiment input features as the proposed approach.

Looking at the horizontal daily logarithmic return performance comparisons between the proposed approach and other benchmark forecasting approaches as shown in Table 6, it is clear that when our proposed approach is applied to a higher frequency Bitcoin price dataset, it is still able to significantly outperform the other four benchmark forecasting models in terms of RMSE, MAPE, MAE and DA across all three forecasting horizons. As a result, the validity of our proposed approach is further verified. The results show that our proposed approach is able to effectively improve the Bitcoin market predictive ability of the model.

Table 6 Higher frequency price performance comparison

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Trading results comparisons

To determine the practicality of using the proposed model as a decision support tool in real world Bitcoin trading, algorithmic trading is conducted based on the predicted Bitcoin market price. The forecasting model produces a buy signal if the predicted Bitcoin market price next day will be higher than the current Bitcoin market price today. If the predicted price next day is lower than the price today, the forecasting model generates a sell signal. Otherwise, the model produces a hold signal. The model will then use the generated signals to conduct algorithmic trading. Specifically, if a buy signal is generated, all the available capital will be used to purchase Bitcoin at that specific time. On the other hand, if a sell signal is generated, all the purchased Bitcoins will be *sold* at that specific time. In this study, the initial investment capital is set to \$100,000.

To be potentially useful as a trading decision support system, the forecasting model must have a DA higher than 50%, which could be obtained by chance. In this paper, we set the DA

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The out-of-sample testing interval consists of 280 trading days from March 28, 2020 to January 1, 2021. As illustrated in Fig. 3, the Bitcoin market price experienced significant fluctuations during the testing interval. The out-of-sample testing period is split into two different intervals—the “Up” interval consisting of 93 days from March 28, 2020 to July 11, 2020, and the “Down” interval consisting of 159 days from July 12, 2020 to January 1, 2021.

Figure 7 and Tables 7 and 8 present the annualized returns obtained by all trading strategies in the out-of-sample “Up” interval, “Down” interval, as well as the overall interval. The results clearly demonstrate that the proposed VMD-LMH-BiLSTM prediction model outperforms the naïve buy-and-hold strategy, as well as other benchmark models across all testing periods. Specifically, during the “Up” interval where the Bitcoin market price soared quickly, an investor trading on the Bitcoin market using signals generated by the proposed VMD-LMH-BiLSTM model, achieves an annualized return of 383.06% after 93 trading days. This is a 58.67% increase compared to the buy-and-hold strategy. During the “Down”

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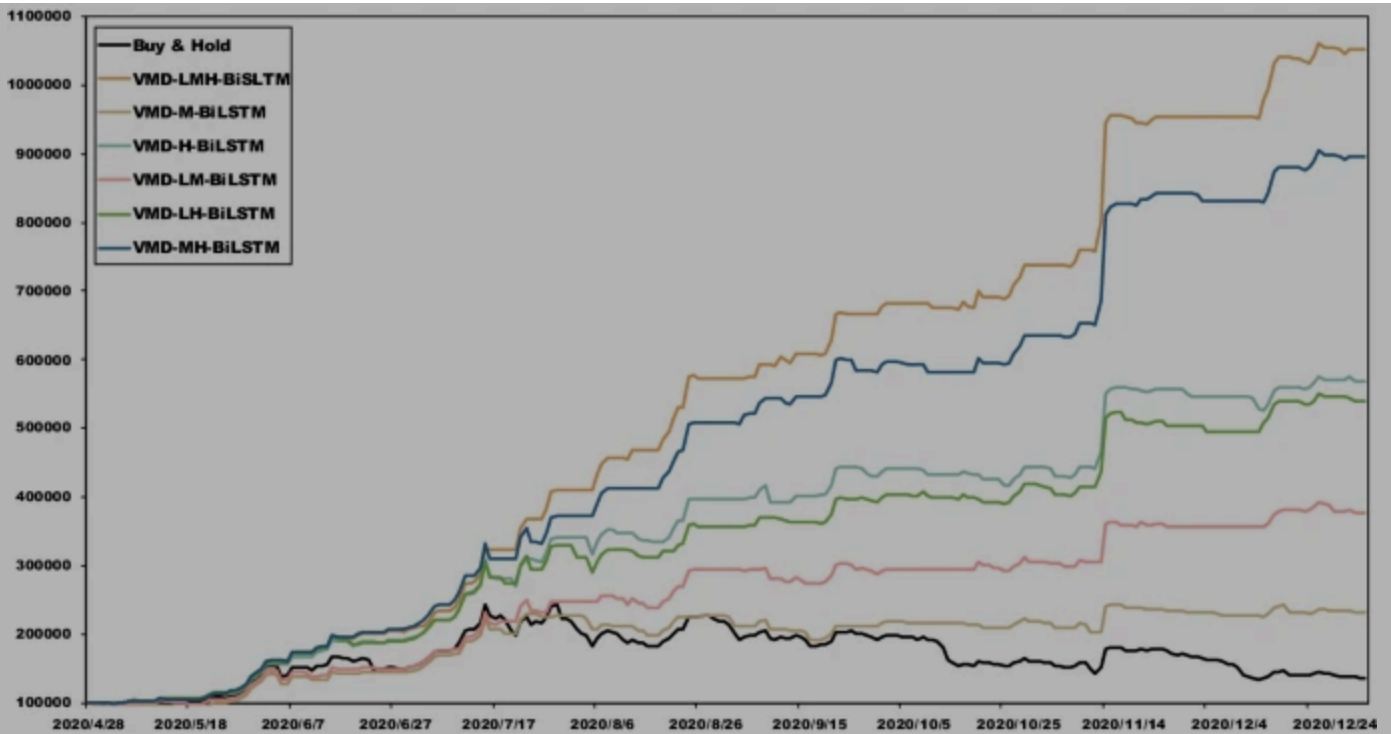
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Trading performance of all trading strategies

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frequencies that exist in the time series. In addition, by adopting a bidirectional neural networks structure, the proposed model is able to effectively capture the two-way, sequential relationship within the time series.

According to our empirical results, the proposed VMD-LMH-BiLSTM model outperformed all the benchmark models in terms of prediction accuracy across multiple forecasting periods. Moreover, the proposed model also displays superior trading performance when compared to other benchmarks such as the buy-and-hold strategy. In particular, our model shows strong consistency in generating profits under volatile market conditions. It effectively reduces the negative impacts of bull markets, which is especially important for avoiding losses in the highly volatile Bitcoin market. Overall, the superior performances demonstrated by the proposed VMD-LMH-BiLSTM model, in terms of prediction accuracy and trading results, indicate that it is not only an effective prediction model but also a potentially useful trading support system.

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Availability of data and materials

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

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Contributions

YL and SJ have analyzed data, drafted the work and implement software. XL have designed the work and substantively revised the manuscript. SW have made substantial contributions to the conception. All authors read and approved the final manuscript.

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Ethics declarations

Competing interests

The authors declare that they have no competing interests.

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